Fast Underwater Image Enhancement for Improved Visual Perception

Indian Institute of Technology Jammu

भारतीय प्रौद्योगिकी संस्थान जम्मू
INDIAN INSTITUTE OF TECHNOLOGY JAMMU
विद्याधनं सर्वधन प्रधानम्

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Submitted To: Dr. Badri Narayan Subudhi

Submitted By: Aditi Gautam, Tushar

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Overview of the Topic

- **FUnIE-GAN**: Fast underwater image enhancement using conditional GANs
- **EUVP Dataset**: 20K underwater images for training
- **Performance & Feasibility**: Real-time processing for underwater robotics

Research Papers Used

- **Paper 1: Underwater Image Enhancement Using Deep Transfer Learning**
  Yunfeng Zhang, Qun Jiang, Peide Liu, et al.
  IEEE Journal of Oceanic Engineering, 2023
  DOI
- **Paper 2: Fast Underwater Image Enhancement**
  Md Jahidul Islam, Youya Xia, Junaed Sattar
  IEEE Robotics and Automation Letters, 2020
  arXiv

Background and Motivation

- Traditional methods need extra scene data and are too slow
- Unique underwater optics (blue/green dominance)
- Limited real-world paired datasets in deep learning

Objectives

- Real-Time Enhancement: Develop FUnIE-GAN model
- Dataset Creation: EUVP dataset with 20K images
- Performance Boost: Improve object detection tasks

Paired Images: FUnIE-GAN Architecture

- **Generator Architecture (FUNIE-GAN):**
  - U-Net + Skip Connections: Encoder-decoder design with skip connections to preserve spatial details.
  - Efficient Down/Up-Sampling: Uses MaxPooling (down) and UpSampling+Conv2D (up) for resolution recovery.
- **Discriminator (FUNIE-GAN):**
  - PatchGAN: Analyzes 16x16 patches for local realism (low computational cost).
  - Conditioned Feedback: Processes paired (real/fake + distorted) inputs to guide generator training.

FUnIE-GAN Losses (Discriminator Loss)

- **Adversarial Loss (Discriminator Loss):**
  - Adversarial Loss is used to train the discriminator D to distinguish between real and fake images.
  - **Formula:**

$$\underbrace{\mathbb{E}_{x\sim p_{\mathbf{data}}}\left[(D(x)-1)^2\right]}_{\text{Loss for real images}}+\underbrace{\mathbb{E}_{z\sim p_z}\left[D(G(z))^2\right]}_{\text{Loss for fake images}}$$

FUnIE-GAN Losses (Generator Loss)

- **MAE (L1) Loss**
  Ensures pixel-level accuracy between generated and real images.
  Formula:

$$L_{\text{MAE}} = \mathbb{E}_{x,y}\left[\|y - G(x)\|_1\right] = \frac{1}{HW}\sum_{i,j}|y_{i,j} - G(x)_{i,j}|$$

- **Content Loss (VGG19-based)**
  Maintains high-level feature similarity using VGG19 activations.
  Formula:

$$L_{\text{content}} = \mathbb{E}_{x,y}\left[\|\phi(y) - \phi(G(x))\|_2^2\right]$$

  where ϕ denotes VGG19's `block5_conv2` features.

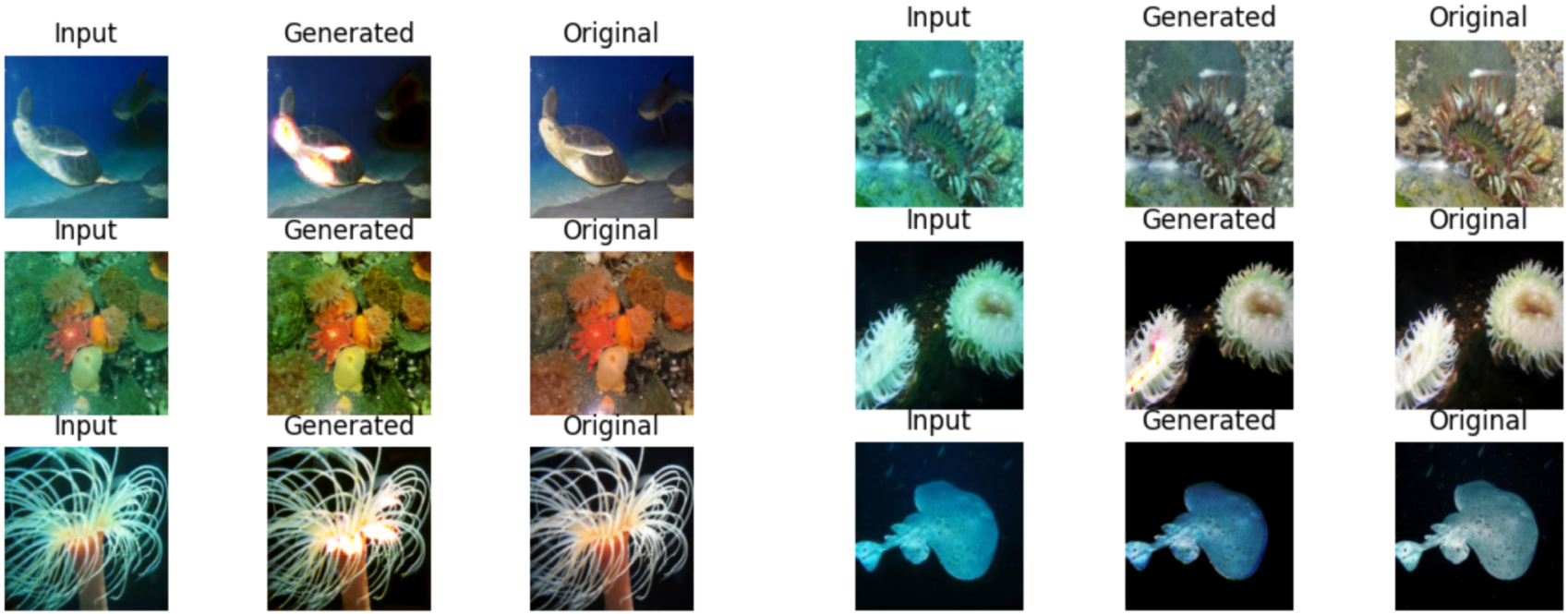
- **Combined Generator Loss**
  Balances pixel-level and perceptual quality with weighted terms.
  Formula:

$$L_{\text{total}} = 0.7L_{\text{MAE}} + 0.3L_{\text{content}}$$

Training Model

- **Setup:** Trains FUnIE-GAN on paired EUVP data with 200 epochs, batch size 4, validation interval 2000.
- **Training:** Alternates discriminator and generator updates; logs losses every 50 steps.
- **Architecture:** Combines adversarial and reconstruction losses in a GAN framework.

Generated Enhanced Image



Enhanced Testing

- **Original Testing Script - Key Points:**
  - Single-Image Testing: Processes images individually, missing batch optimization.
  - Basic Timing Metrics: Only measures total time/FPS, no quality (PSNR/SSIM) evaluation.
- **Enhanced Testing Features:**
  - Batch Processing: Tests multiple batch sizes (e.g., [1, 4, 8, ..., 46]) to find the optimal balance between speed (FPS) and quality.
  - Comprehensive Metrics: Calculates PSNR and SSIM for each batch and reports FPS, time/batch, and averages across runs.
  - Automated Optimization: Dynamically determines the optimal batch size (optimal_bs) by balancing FPS and SSIM.

Testing Output

```
Model loaded successfully

Processed 46 images in 32.27 seconds
Average speed: 1.43 FPS
Outputs saved to: ./data/output/
```

Original Testing

```
Performance Summary for batches:
Batch    FPS      Time       PSNR       SSIM
1        2.70     0.3710     33.64      0.8280
4        5.15     0.7774     33.72      0.8199
8        7.83     1.0211     33.89      0.8188
16       8.30     1.9276     33.83      0.8168
24       9.21     2.6066     33.85      0.8168
32       11.63    2.7522     33.87      0.8168
36       11.38    3.1642     33.86      0.8170
40       15.63    2.5593     33.81      0.8159
44       26.55    1.6570     33.95      0.8149
45       36.20    1.2432     34.18      0.8158
46       8.41     5.4671     33.84      0.8182

Optimal batch size: 45
(Balancing speed and image quality)
Outputs saved to: ./data/output/
PS C:\Users\ADITI\Desktop\funiegan>
```

Enhanced Testing

References

- Islam, M.J., et al. (2020). IEEE RA-L 5(2)
- Zhang, Y., et al. (2023). IEEE J. Oceanic Eng. 48(2)